



DPUSGERoundGauge

Overlays a view of SILAB's visualization SGE with a round gauge. The round gauge is displayed as a HUD (head up display) element, two-dimensionally.

This DPU can be used together with DPUSGEInstrumentPanel to overlay the view with a car dashboard.

1. Inputs:

<i>Show</i>	Specifies if the round gauge is currently visible (<i>true</i>) or invisible (<i>false</i>). Default: <i>true</i> .
<i>Value</i>	Current value of the input parameter.

2. Dynamic parameters for the background:

<i>X, Y</i>	Position of the round gauge in SILAB's overlay coordinate system
<i>Size</i>	Width and height of the round gauge, in pixels (the instrument is always quadratic)
<i>Z</i>	Z position of the round gauge (0..10, where 10 is "in front" or "on top" of all other scene elements)

3. Static parameters for the background:

<i>TexInstrument</i>	Resource name of the background texture (e.g. " <i>textures.hud.speedometer</i> ")
<i>Visibility</i>	Visibility category for the object. It can be used to display the object on one of the views updated on a single graphics computer only.

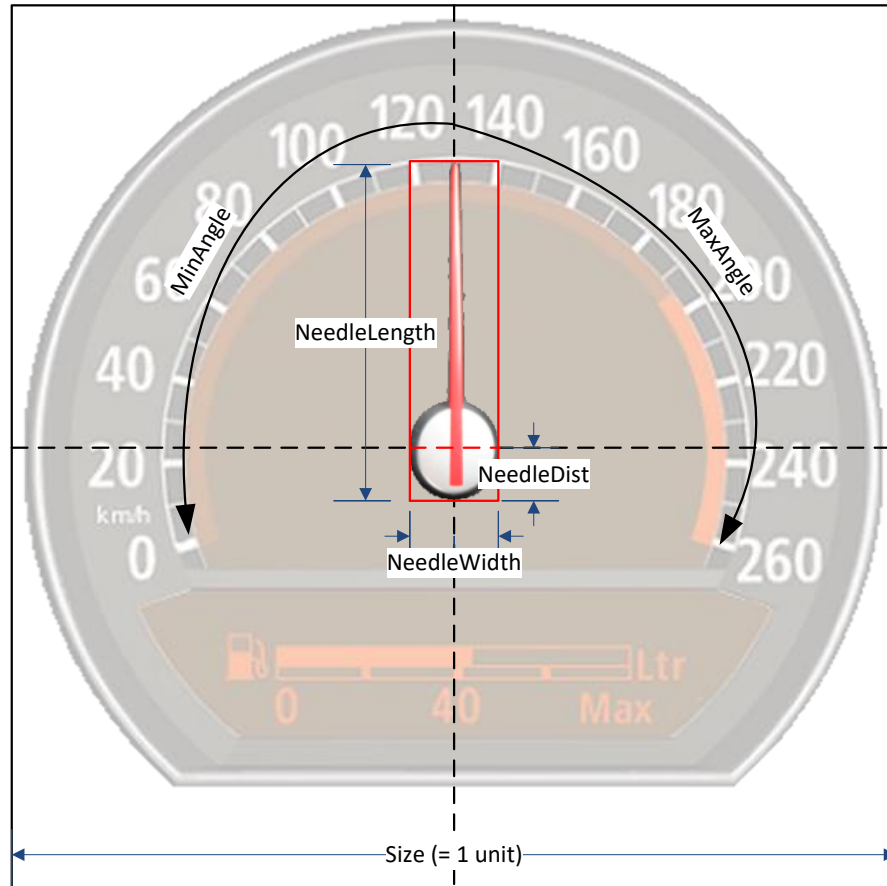
4. Static parameters for the needle:

<i>TexNeedle</i>	Resource name of the needle texture (e.g. " <i>textures.hud.needle</i> ")
<i>NeedleDist</i>	Distance of the needle mounting point to its end (relative to total gauge size)
<i>NeedleLength</i>	Length of the needle (relative to total gauge size)
<i>NeedleWidth</i>	Width of the needle (relative to total gauge size)
	See the figure below for the meaning of these parameters.
<i>MinAngle</i>	Minimal angle [rad], where 0 is pointing upwards
<i>MaxAngle</i>	Maximal angle [rad], where 0 is pointing upwards
<i>MinValue</i>	The minimum displayable value: For this <i>Value</i> , the needle assumes the angle <i>MinAngle</i> .
<i>MaxValue</i>	The maximum displayable value: For this <i>Value</i> , the needle assumes the angle <i>MaxAngle</i> .



5. Explanation of needle parameters:

The needle parameters are explained in the following figure.



6. Dependencies to other DPUs:

DPUSGERoundGauge must run together with **DPUSGEWorld** and **DPUSGEView**. The Index values of the DPUs must be set according to the following relation:

$SGEWorld.Index < SGERoundGauge.Index < SGEView.Index$

7. Example:

The following script implements a speedometer:

```
DPUSGERoundGauge Speedometer
{
    Computer = {VIS};
    Index = 50;

    TexInstrument = "hud.bg_tacho";

    TexNeedle = "hud.needle";
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
NeedleWidth = 0.1094;  
NeedleLength = 0.38;  
NeedleDist = 0.0547;
```

```
MinAngle = -1.9134;  
MaxAngle = 1.9134;
```

```
MinValue = 0;  
MaxValue = 260;
```

```
X = 236;  
Y = 270;  
Z = 8.45;
```

```
};
```

With an input parameter (*Value*) of 80, the following display results:

